

An Efficient EVM Based Jamming Detection in 5G Networks

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Abstract—5G is expected to support high data rate, low latency and massive number of connectivity requirements. Due to these features, 5G will serve many different tasks such as increased reality, internet of things, emergencies and remote interventions, besides civil communication. Therefore, it has become more important to detect malicious jamming attacks in 5G. Signal level and error rate metrics have been widely proposed for jamming detection in various wireless networks. However, most of the metrics are insufficient to detect jamming at high SJR (Signal-to-Jamming Ratio) values. On the other hand, for 5G networks, we propose a more sensitive and informative jamming detection method that exploits the "EVM-vs-RB" measurement obtained by calculating EVM (Error Vector Magnitude) metric for each RB (Resource Block). The presence of jamming is then detected when the predetermined threshold level for EVM is exceeded on the proposed measurement set. As the EVM information is obtained for each RB, it is also revealed which RBs, that is frequency bands, are subject to the jamming attack. Obtained simulation results show that jamming attacks, which cannot be detected in high SJR situations with the classical metrics, can be detected by the proposed method even at high SJR values such as 20 dB.

Index Terms—5G, Jamming Detection, EVM, Resource Block

I. INTRODUCTION

5G is a new generation technology that aims to provide wireless network connections and many standart applications in a faster and more practical way. Higher data transmission speed, increased capacity, reduced communication delay are among the best-known new features of 5G. Thanks to these features, 5G is expected to serve many critical applications such as connected vehicles, virtual reality, smart homes and remote intervention. It is therefore quite important to be cautious against malicious jamming attacks in these networks. For this reason, a jamming detection method with high accuracy and sensitivity and low complexity is needed. Furthermore, the ability of the jamming detection method to provide the frequency band information that the jammer occupies in the spectrum is critical for assigning the correct frequency resources to UEs (User Equipments) in 5G. Considering all these factors, for jamming detection in 5G networks, we propose the EVM-vs-RB measurement which is capable of providing the mentioned requirements.

There are many jamming detection methods proposed for centralized and decentralized wireless networks, GNSS, and satellite communication systems. Signal level and error rate metrics are benefited in most of the methods. The effect of selected metrics on detection accuracy has been discussed widely in the literature [1]-[3].

Comparing RF power of the received signal with a threshold level is a quite common method for jamming detection [4]-[6]. For example, the RF power measured in time domain is utilized for GNSS jamming detection in an experimental setup [7], but it is observed that jamming detection probability is decreased when the SJR is greater than 5 dB. On the other hand, in [8], RF power is measured in frequency domain and used for jamming detection in frequency hopping systems. This study also presents the spectrum information about jamming, but operation complexity of the study is quite high (greater than $O(N^2)$) compared to $O(N)$ which is the complexity of our proposed work.

Many authors have analytically derived the optimum energy thresholds by calculating the receiver statistics. Firstly received signal expressions for the hypotheses of the presence and absence of the jammer are derived analytically according to transmitter, receiver and channel model parameters, and then likelihood functions of the hypotheses are calculated with the signal expressions. Finally, optimum jamming detection thresholds are obtained as a result of likelihood tests. Such jamming detection methods are widely proposed for many areas, e.g., cellular communication systems [9], [10], mobile ad-hoc networks [11], wireless sensor networks [12], SIMO systems [13], Frequency Hopping/Binary Frequency Shift Keying systems [14], DSSS systems [15], satellite communication systems [16] and cognitive radio networks [17]. Although such studies are focused on jamming detection accuracy for different communication systems, they are not concerned to provide jamming signal type, frequency and bandwidth information which are vital importance in terms of the measures that can be taken against the jammer.

A lot of metrics such as RSSI (Receiving Signal Strength Indicator), PDR (Packet Delivery Ratio), CER (Chip Error Rate), SNR (Signal to Noise Ratio), BPR (Bad Packet Ratio), CST (Carrier Sensing Time) and CBR (Channel Busy Ratio) are also proposed for jamming detection. In some studies,

jamming is detected with threshold levels set on these metrics [18]-[26]. In addition, some authors track these metrics statistically and explain any anomalies by the presence of the jammer [27], [28]. On the other hand, in [29], PDR and RSS metrics were provided as input to SVM (support vector machine), neural network and random forest algorithms for jamming detection, and then these machine learning algorithms were compared in terms of detection accuracy. The metrics are also considered as input of the machine learning algorithms used in [30] and [31]. As with similar studies, these studies mainly focus on jamming detection accuracy and do not inform us about jamming signal characteristics.

Unlike the literature studies that propose to use metrics such as RF power and data packet error for jamming detection, the EVM-vs-RB measurement is proposed to be used for jamming detection in 5G networks in this study. By this way, higher sensitivity, lower complexity together with providing useful spectrum information about jammer are also able to be obtained. The EVM metric which will be explained in Section III-B is an error magnitude measure for IQ symbols and it is obvious that the magnitude of the error increases in case of jamming attack. A RB, on the other hand, is the minimum frequency band unit assigned to a UE in LTE and 5G. In this study proposal, the EVM values calculated for each RB are compared with a predetermined upper limit threshold, and then which RBs are under the jamming attack is detected by checking whether the threshold level is exceeded. Since each RB represents different frequency sub-band, spectral characteristic of the jammer is revealed by the EVM-vs-RB (obtained only with $O(N)$ complexity). Simulation results obtained for different jamming signal types in Section V shows that the capability of the proposed method.

Since throughput, PDR and BER (Bit Error Rate) performance do not get worse at high enough SJR values, it is not possible to detect the presence of jammers by these metrics at the high SJR values. On the other hand, because the EVM changes can still be perceived at high SJR values, for example 20 dB, the jamming is able to be detected by the proposed method even if there is no data packet error. Success of our work on jamming detection sensitivity is shown for the high SJR situations where data transmission achievement is %100 in the simulation results in Section V.

II. JAMMER MODELS

A. Tone Jammers

This class of jammers includes the simplest and most common jammer types. Tone jammers discussed in this work refers to "Single-Tone" and "Multi-Tone" sinusoidal signals. The tone jammers can be modeled by (1).

$$j(t) = \sum_{k=1}^K \sqrt{P_k} \exp(j(2\pi f_k t + \theta_k)) \quad (1)$$

Where,

- $k = 1, 2, \dots, K$ represents the tones
- P_k is power

- f_k is frequency
- θ_k is phase

B. Chirp Jammers

A chirp jamming signal sweeps its frequency from low frequency to high frequency (or vice versa) over a certain period of time. After the sweep process is completed, it is restarted at the initial frequency. These jammers can be expressed by (2).

$$j(t) = \sqrt{P_j} \exp(j(f_q(t) + \theta_j)) \quad (2)$$

Where,

- P_j is jamming power
- $f_q(t)$ is the instantaneous frequency indicated in (3)
- θ_j denotes the initial phase of the jammer

$$f_q(t) = 2\pi f_j t + \pi b((f_{max} - f_{min})/T_{swp})t^2 \quad (3)$$

Where,

- f_j starting frequency of the sweep (at time T_{swp})
- f_{min} the minimum frequency sweep
- f_{max} the maximum frequency sweep
- T_{swp} is the sweep period
- $b = \pm 1$ is a flag determining if we have an up-chirp ($b = 1$) or a down-chirp ($b = -1$).

III. JAMMING DETECTION METRICS

A. PDR (Packet Delivery Ratio) and Throughput

Malicious jamming attacks may aim to make data transmission impossible or to reduce the amount of successfully transmitted data in 5G, as in other wireless communication systems. The PDR metric expressed in (4) has been proposed by a significant majority in the literature for jamming detection, as it tells the ratio of data packets transmitted without fail. The equivalent of this metric in 5G is "Throughput" which is the amount of data moved successfully from transmitter to receiver in a given time period. It is usually measured in percentages or bits per second (bps).

$$PDR = \frac{\sum \text{Number of packets received}}{\sum \text{Number of packets sent}} \quad (4)$$

B. EVM, Error Vector Magnitude

Many digital modulation schemes (e.g., M-PSK, M-QAM) are based on the IQ symbols which have certain amplitude-phase values and represents unique bit sequences. For example, the placement of the symbols belongs to 16-QAM modulation in the constellation diagram is shown in Fig. 1. Each of the symbols represents a matchless pattern of 4 bits.

However, in practice, disruptive factors such as jamming, low SNR, inter-symbol interference, multipath fading and phase noise of RF components corrupt the distribution of the IQ symbols in the constellation diagram. In this work, we are only concerned with jamming-induced distortions. Locations of the received symbols in the constellation diagram will deviate a certain amount depending on jamming power.

EVM is a way to quantify the distance between the received (distorted) symbol location and the correct symbol location. The calculation of the EVM is illustrated in Fig. 2.

EVM is calculated for all IQ symbols separately as expressed in (5). Since each of the N symbols is used once for vectoral difference calculation, the EVM has $O(N)$ complexity. Depending on the modulation type, IQ symbols with EVM values above certain levels cannot be demodulated correctly and consequently bit errors arise.

Since, RMS EVM (6) is directly proportional to the BER and maximum EVM (7) has the ability to reveal instantaneous distortions in the received signal, both of the measurements are applicable for jamming detection. The maximum EVM is capable of detecting jamming attacks that take a shorter time than the legitimate signal time thanks to the aforementioned ability of reveal abrupt distortions in the received signal. Therefore the maximum EVM has been proposed in this study.

$$EVM_k(\%) = 100 \sqrt{\frac{e_k}{\frac{1}{N} \sum_{k=1}^N (I_k^2 + Q_k^2)}} \quad (5)$$

$$EVM_{RMS}(\%) = \sqrt{\frac{\sum_{k=1}^N EVM_k^2}{N}} \quad (6)$$

$$EVM_{MAX}(\%) = \max_{k \in [1, \dots, N]} EVM_k \quad (7)$$

Where,

- $e_k = (I_k - \tilde{I}_k)^2 + (Q_k - \tilde{Q}_k)^2$
- I_k and Q_k represent ideal in-phase and quadrature values of the k^{th} symbol.
- $\tilde{I}_k$ and $\tilde{Q}_k$ represent received in-phase and quadrature values of the k^{th} symbol.
- N is the input vector length.

IV. SYSTEM MODEL

A 5G downlink data transmission setup was installed in MATLAB environment [32]. In order to examine jamming effects; multi-tone and chirp jamming signals are considered

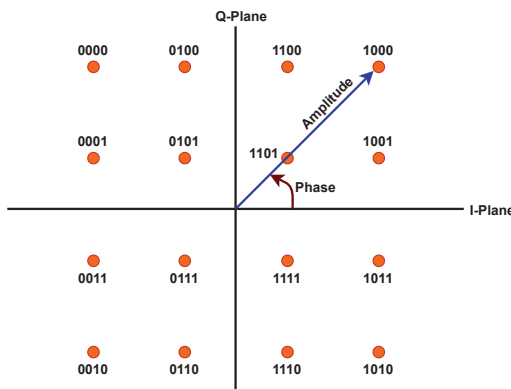


Fig. 1: 16-QAM Constellation (Correct Positions of the IQ Symbols)

in the simulations. The throughput and proposed EVM-vs-RB which are candidate metrics for jamming detection, are calculated for a UE that is subjected to the jamming attack.

In the downlink data transmission, the gNB (base station) creates the data packet, applies error detection and channel coding to this packet, modulates it to the RF band, and finally transmits it to the UE via MIMO antennas. However, in addition to the legitimate signal sent by the gNB, the UE also receives jamming signals. All these stages are shown in more detail in Fig. 3.

In the DL-SCH (Downlink Shared Channel) stage [33], raw user data is encoded for error detection and error correction by CRC (Cyclic Redundancy Check) and LDPC (Low-Density Parity-Check Code) algorithms respectively [34]. Encoded codewords are then transferred to the PDSCH (Physical Downlink Shared Channel) stage.

PDSCH [35] is the physical downlink channel that carries user data. The codewords are first scrambled so that unauthorized devices cannot decode the broadcast, then modulated to IQ (In-phase, Quadrature) domain. IQ modulation options [36] can be QPSK, 16-QAM, 64-QAM and 256-QAM. Depending on quality of communication channel, higher degree modulation types can be used in order to increase the communication speed.

In 5G, MIMO antenna structures are utilized to transmit signal over the wireless environment. The layer mapping stage distributes the IQ symbols to layers. The increased number of layers means that parallel data streams can be transmitted simultaneously via MIMO antennas, which means that the data transmission rate will increase proportionally. Increasing the number of layers also depends on the goodness of the communication channel. By the antenna precoding stage, it is determined to which antennas the symbols in the layers will be mapped. In this stage, reference IQ symbols called DM-RS (Demodulation Reference Signals) [36] are also added to the original symbols so that the UE can estimate the channel state.

The symbols mapped to the antennas are modulated to RBs (Resource Blocks) in the RF band by OFDM. The frequency band assigned to an UE for data transmission in 5G is expressed by number of RBs. One RB corresponds to 12 sequential OFDM subcarrier frequencies [36]. Finally, the RF signal is transmitted to the UE via the MIMO antennas. Until

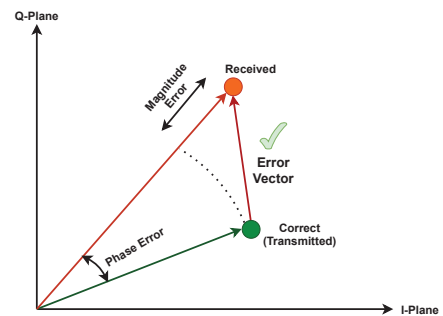


Fig. 2: Visual Representation of EVM Measurement

the signal reaches the UE, it is subject to multipath fading, additive white noise and jamming effects.

In UE receiver terminals, the IQ symbols carried by the assigned RBs are obtained after OFDM demodulation. However, the fading and jamming effects on the RF signal are transferred to the IQ symbols by the OFDM demodulation. In order to eliminate such disruptive channel effects on the IQ symbols, equalization process according to MMSE criteria is applied to the IQ symbols with the channel response information which is estimated by using the DM-RS symbols. The fading effect is minimized thanks to the equalizer, but the jamming effect is still observed on the IQ symbols as shown in the results in Section V. Therefore, the EVM-vs-RB measurement taken with equalized IQ symbols shows the jamming effect more clearly and is proposed for jamming detection in this study.

Afterwards, the symbols are subjected to QPSK or QAM demodulation to obtain corresponding code words. Code words are then converted into bit strings by the LDPC decoder. After the bit strings are obtained, metrics such as throughput, PDR and BER, which are widely proposed for jamming detection in the literature can be calculated. Whereas, since the proposed EVM-vs-RB is measured with the IQ symbols in the RBs, the jamming is caught in an earlier stage without need for IQ demodulation and decoding. Also, at sufficiently high SJR situations, it is not possible to detect jamming with the throughput, PDR and BER metrics, as the data transmission achievement will be %100. On the other hand, it is shown in the results in Section V that jamming can still be detected at the high SJR situations by the proposed method.

V. EXPERIMENTAL RESULTS

The simulations have been carried out according to the setup parameters in Table I. The EVM-vs-RB measurements have been taken for both unequalized and equalized IQ symbols. As given in Fig. 4b-7b, the EVM-vs-RB values reduce after the equalization process.

In Fig. 4, the spectrum of the received signal and the EVM-vs-RB are given for the jammer-free case. An expected fluctuation in the spectrum due to the multipath channel effect have been observed in Fig. 4a. It is quite remarkable

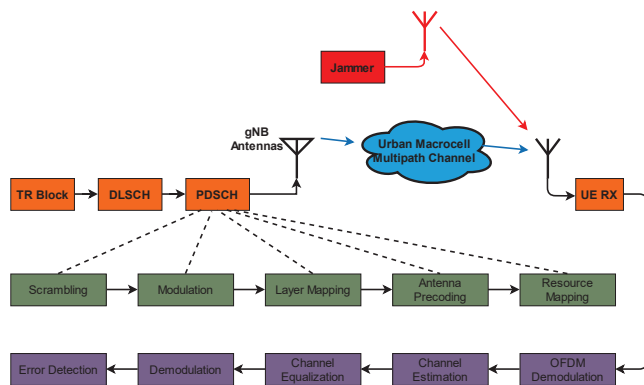


Fig. 3: 5G Downlink Data Transmission

TABLE I: Downlink Data Transfer Terms

Parameter Name	Value	Explanation
OFDM Subcarrier Spacing	30 kHz	
RBs	51	Corresponds to a bandwidth close to 20 MHz with 30 kHz SCS
IQ Modulation Type	16-QAM	
Channel Delay Profile	CDL-C	Urban Macrocell Model, NLOS

that the EVM-vs-RB graph (Fig. 4b) taken with unequalized symbols can capture the frequency-dependent change in the spectrum exactly. How the equalization process minimizes the fluctuation in the spectrum, that is, the fading effect, is also shown with the EVM-vs-RB graph taken with the equalized IQ symbols (Fig. 4b).

These measurements have also been taken for presence of the jammers. The results are given in Fig. 5-7. Both the fading and jamming effects are seen in the EVM-vs-RB measurements taken with the unequalized symbols. After the equalization, the fading effect is minimized, but the jamming effects are still observed clearly in the measurement results. Therefore, we propose to use EVM-vs-RB measurements taken with equalized symbols for jamming detection.

In Fig. 8 and 9, we have presented the EVM-vs-RB and throughput measurements taken under multi-tone jamming. It has been shown in the EVM-vs-RB graphs that jamming detection can be easily done with a single threshold level. Moreover which RBs, that is frequency bands, are affected by the jamming attack can be clearly detected by the EVM-vs-RB. In order to demonstrate jamming detection sensitivity of the proposed method, more measurements with different SJR values have been achieved are given in Fig. 8 and 9.

As seen in Fig. 8d and 9d, the throughput becomes %100 when the SJR value is greater than 10 dB. In this case, it becomes impossible to detect jamming by throughput observation. On the other hand, the fact that jamming effect can be detected clearly by the EVM-vs-RB for 20 dB SJR value (Fig. 8c and 9c), reveals the success of the proposed method in terms of sensitivity. Fig. 10 shows that the sensitivity skill of the proposed method also applies to chirp jammers.

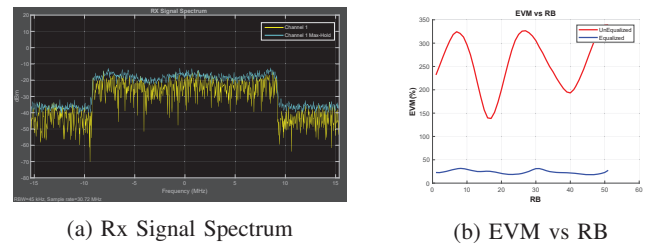
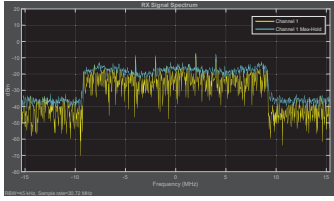
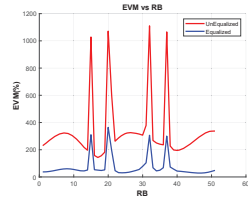


Fig. 4: No Jammer Case

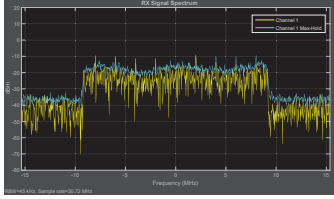


(a) Rx Signal Spectrum

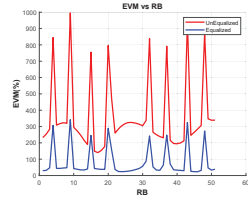


(b) EVM vs RB

Fig. 5: Multitone Jammer Case (SJR = 0 dB)

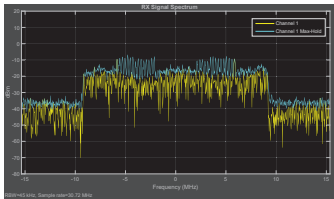


(a) Rx Signal Spectrum

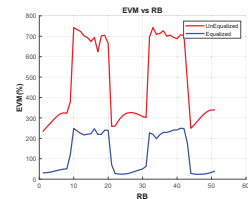


(b) EVM vs RB

Fig. 6: Multitone Jammer Case (SJR = 0 dB)

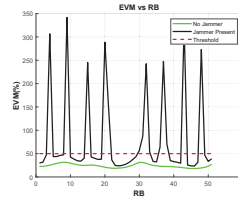


(a) Rx Signal Spectrum

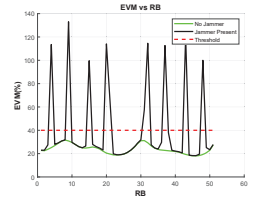


(b) EVM vs RB

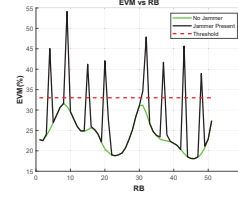
Fig. 7: Partial Bant Chirp Jammer Case (SJR = 0 dB)



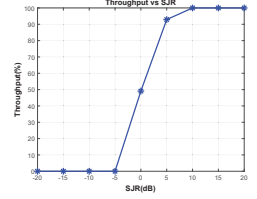
(a) SJR = 0 dB



(b) SJR = 10 dB

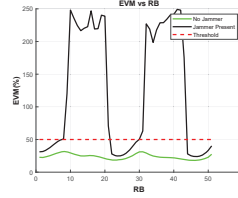


(c) SJR = 20 dB

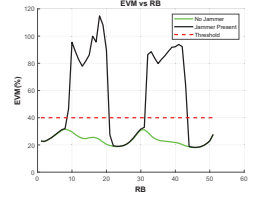


(d) Throughput vs SJR

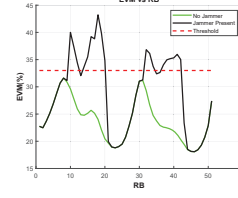
Fig. 9: EVM and Throughput Measurements for Multitone Jammer



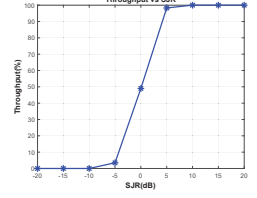
(a) SJR = 0 dB



(b) SJR = 10 dB



(c) SJR = 20 dB



(d) Throughput vs SJR

Fig. 10: EVM and Throughput Measurements for Partial Bant Chirp Jammer

VI. CONCLUSIONS

In this work, we proposed an efficient jamming detection method that considers the EVM metric for each RB. Higher sensitivity, lower complexity together with to provide useful spectrum information about jammer are the main contributions. This method is suitable for LTE, 5G and perhaps will also be convenient next generation wireless communication systems that will use the RB architecture. Experimental results showed that the proposed method can detect different jammer types successfully, even at low jamming powers where the throughput approaches to %100. In addition, in which frequency bands are confronted to the jamming attack is also be able to determined and we believe that this information can be very useful for anti-jamming spectrum scheduling works.

Fig. 8: EVM and Throughput Measurements for Multitone Jammer

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